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A robust method for critical micelle concentration determination using coumarin-6 as a fluorescent probe†

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The critical micelle concentration (CMC) is the point at which micelles are self-assembled from surfactants added to the system. Determination of the CMC is commonly based on surface tension and conductivity measurements since these are relatively simple and well described. The dye micellization method performed using pyrene as a fluorescent probe, is limited like other methods by inaccuracy and the lack of a standard procedure. We established a simple, robust, and straightforward method based on dye micellization, using coumarin-6 as a fluorescent probe for CMC determination. We first optimized the method and demonstrated the problematic pyrene detection by a plate reader. We compared the novel method to the established surface tension method using anionic, cationic and non-ionic surfactants, and compared to reported values in the literature. We selected to measure sodium dodecyl sulfate using the conductivity method, followed by visualization by fluorescence microscopy. The values obtained by our method were reproducible and in accordance with the values reported in the literature, regardless of the ionic charge of the surfactant. Adopting such a robust protocol would be beneficial for pharmaceutical research and industry and has the potential to replace the more laborious and less accurate commonly available methods.

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Introduction

The critical micelle concentration (CMC) is defined as the concentration of amphiphilic molecules above which micelles start to self-assemble to form a liquid colloidal solution. Surfactants are amphiphilic molecules composed of hydrophobic tails and usually by hydrocarbonic and hydrophilic heads that can be ionic, non-ionic or zwitterionic. In aqueous solution, surfactants tend to accumulate at the surface, decreasing the interface tension.¹ Above the CMC, saturated amphiphilic molecules accrue at the surface and micelles are detectable in the media assembled from hydrophilic shells and hydrophobic cores, as shown in Fig. 1. The formation of micelles is energetically favorable due to removal of hydrophobic fragments and reforming of hydrogen bonds in the water.² Surfactants have a wide range of applications in industries of personal care and laundry products, agrochemicals, textiles, detergents, cosmetics and pharmaceuticals, food production and processing, petroleum, mineral ores, fuel additives and more.^{3,4}

The determination of CMC is important in pharmaceuticals when dealing with poorly water-soluble drugs or micelles as a targeted carrier.^{2,5} Pharmaceutical companies try to avoid developing drugs that have poor water solubility and discard potential candidates regardless of their possible benefit. The addition of an organic solvent or surfactant to the formulation, for solubility enhancement, exposes the patient to toxicity and side effects.⁶ The usage of micelles as suspended drug carriers can protect the drug from the external surroundings, enhance solubility by 10–5000 fold and prolong the circulation time, but likewise, protect the environment from the drug itself (*e.g.* irritation to the stomach).^{7,8} The CMC value is important in this context since the micelles are less stable below their CMC, resulting in drug precipitation.⁹ Surfactant CMC is also very important in the structural analysis of membrane proteins, their solubilization, isolation, stabilization and crystallization.¹⁰

Various methods have been developed for CMC determination, including measurements of surface tension,^{4,11} conductivity,^{4,12} turbidity,¹³ osmotic pressure,¹³ light scattering^{14,15} and the use of a fluorescent probe,^{16–18} each with its strengths and weaknesses. Surface tension is measured using the DuNouy ring, which is the most standard and acceptable method.^{19,20} Here we present a new simple, straightforward and accurate method of CMC determination. Our method is based on dye micellization using coumarin-6.

Coumarins are a very large and vital family of compounds composed of fused pyrone and benzene rings, with the pyrone

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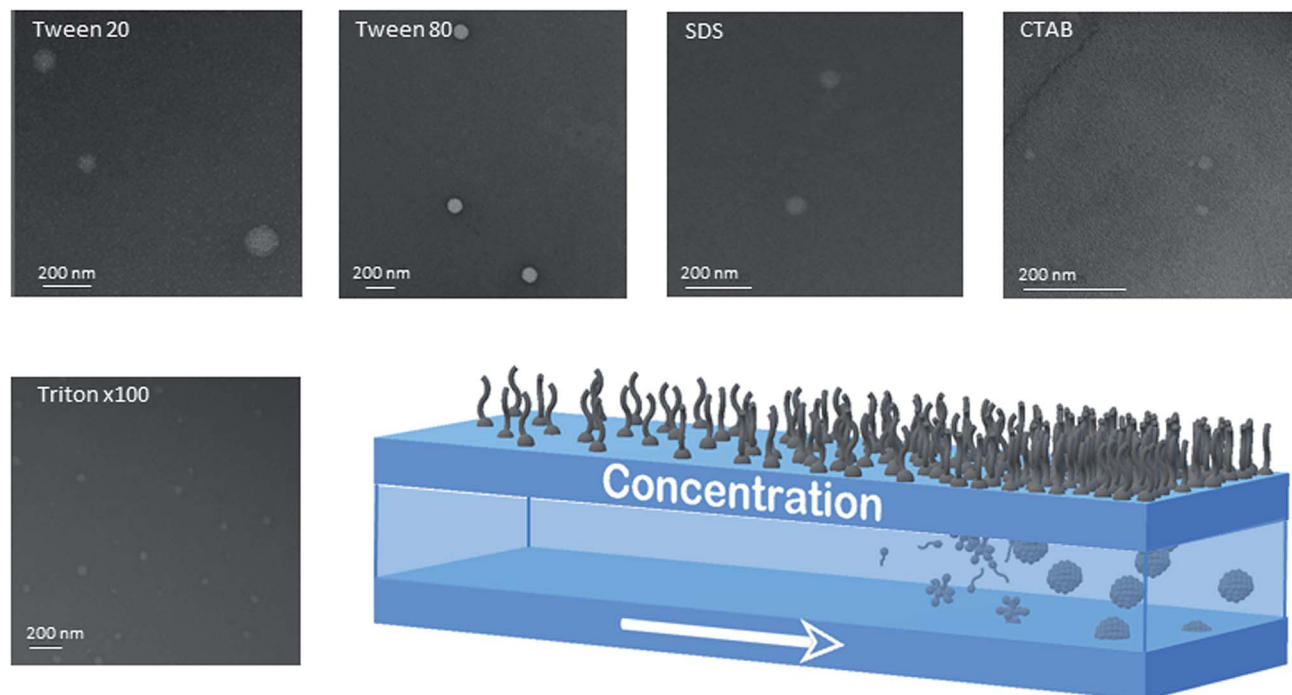


Fig. 1 TEM images of micelles assembled from various surfactants above the critical micelle concentration, incubated with coumarin-6 by the developed method. Tween 20 200 μ M, Tween 80 90 μ M, SDS 25 mM, CTAB 5.5 mM, Triton X100 500 μ M. Also presented is a scheme of micelle formation as the concentration of surfactant at the surface increases. Scale bar 200 nm.

carbonyl group at position two. Due to their high degree of sensitivity to their microenvironment, coumarins are frequently used as sensitive fluorescent probes in diverse systems.²¹ The advantages of using coumarin-6 as a fluorescent probe is its common usage in various applications, including as a common model molecule for drug delivery formulations, its unchanged quantum yield and fluorescence spectrum in different micellar

structures, and the fact that it has no interaction with its microenvironment.^{21–23} The hydrophobicity of the dye and its low water solubility, resolved by the addition of surfactant, make it a perfect candidate for the presented method. Above the CMC, in correlation with the micellization method, the position of the dye inside the hydrophobic core is preferable, and the fluorescence intensity of the solution increases.²⁴ Introducing

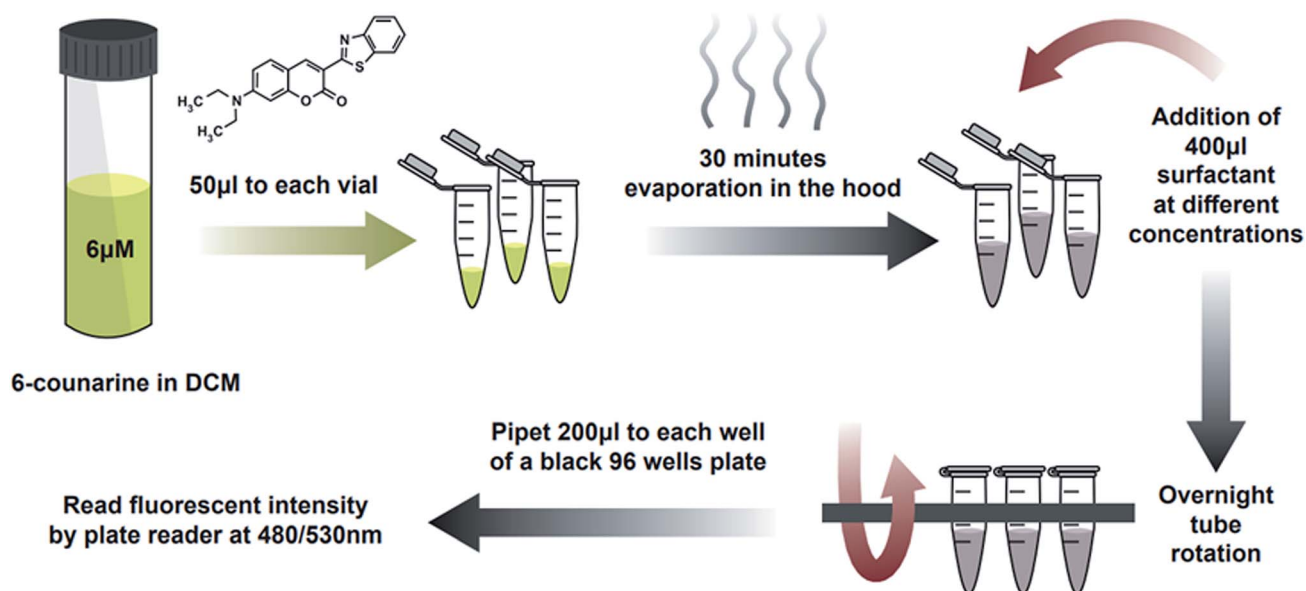


Fig. 2 Illustration of the method depicting step-by-step the procedure presented in the method section using coumarin-6.

such a robust protocol can be of great value for pharmaceutical researchers in academia and industry and may replace the more laborious and less accurate commonly available methods.

Experimental

Materials

Sodium dodecyl sulfate (99%, SDS), hexadecyltrimethylammonium bromide (99%, CTAB), coumarin-6 (98%) and pyrene (99%) were purchased from Sigma-Aldrich Co. (Steinheim, Germany) and polyoxyethylene-20-sorbitans monooleate (TWEEN 80®) was purchased from Fisher BioReagents (New Jersey, USA). Polyoxyethylene-20-sorbitans monolaurate (TWEEN 20®) and octylphenol ethoxylate (Triton X100) were purchased from Avantor (Pennsylvania, USA). Dichloromethane (DCM) was supplied by Bio-Lab (Ashkelon, Israel).

Methods

CMC detection using surface tension measurements. The du Noüy ring method was applied to measure the surface tension by using tensiometer with a platinum ring at room temperature (K20 solid QC tensiometer, KRUSS, Germany). Prior to each test, the ring was rinsed with deionized water and placed over an alcohol burner flame for a short time to clean dirt and liquid remains. Stock solutions of surfactants were prepared by

dissolving the surfactant in double distilled water (DDW) under a gentle stirring for one hour. Working solutions were obtained by diluting stock solutions by desired factor. Each surfactant dilution was vigorously shaken and 20 ml was poured to a glass flask and left for foam disappearance before measuring. Measurements were carried out at least 5 times to represent application of the force required for correct film tearing. CMC was determined by plotting surface tension *versus* concentrations of surfactant.⁴

CMC detection using pyrene as a fluorescent dye. Pyrene was first dissolved in acetone to create a stock solution. Pyrene concentration was controlled by adding an adequate volume from the stock solution to an empty flask and leaving for evaporation before introducing 3 ml of the tested surfactant concentration, to reach a final concentration of 6 μM . Three ml of SDS 5% was added to the evaporated vial and ultra-sonicated (Wisd ultrasonic cleaner, DAIHAN, Korea) for 30 minutes before reading by a spectrofluorometer (FluoroMax4, HORIBA, USA) using 2.5 ml in quartz vials, or by a plate reader using 200 μl per well of a 96-well black plate. Excitation wavelength was set to 336 nm and emission spectrum set to 350–450 nm.²⁵

CMC detection using coumarin-6 as a fluorescent dye. As presented in Fig. 2, a stock solution of 6 μM coumarin-6 in DCM was prepared. Approximately 40 μl from the stock solution was added to an empty minitube (Eppendorf) and left for 30

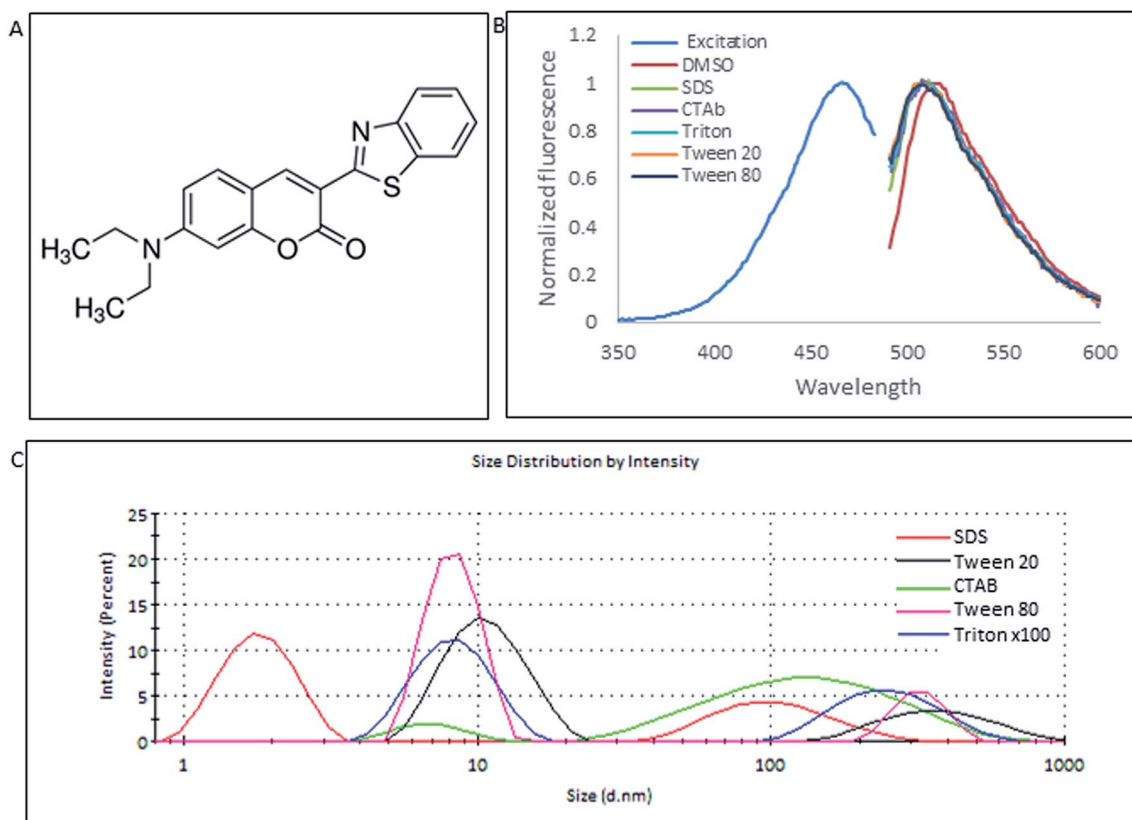


Fig. 3 (A) Coumarin-6 chemical structure and (B) its whole fluorescence spectra of in DMSO and aqueous surfactant solutions above the CMC (0.5 mg mL^{-1}). (C) Hydrodynamic radii changes after different surfactants solutions tested by the method developed, showing two distinctive peaks representing the micelles with or without the encapsulated dye.

minutes to evaporate in the hood. A series of concentrations of several surfactants was made, as detailed above, ranging from none to at least twice the CMC value reported in the literature, and 400 μl from each tested solution was transferred to the Eppendorf containing the solvent evaporated fluorescent dye. All minitubes were left for 20 hours under rotation in the dark at room temperature. Of each solution, 200 μl were transferred to a black 96-well plate and read by a microplate reader at excitation emission wavelengths filter of 485 ± 20 and 528 ± 20 nm respectively (SYNERGY-HT, Biotek, USA). The resultant fluorescence intensities were plotted against the logarithm of the corresponding concentration and the CMC was determined by the intersection of the two tangents created in the graph.

CMC detection by conductometric measurements. The conductivity values of varying surfactant concentrations were measured with a mass spectrometer (AKTA start, GE Healthcare, Sweden). Conductivity *vs.* concentration was plotted and the CMC was obtained by the intersection of the fitting lines above and below the breakpoint.¹²

Visualization of the CMC by fluorescent microscope. Samples of SDS with varying concentrations were imaged using a fluorescent microscope (IX-73, OLYMPUS, Japan), in which fluorescence correlates to dye internalization into the micellar structure. Images were taken using a GFP filter suitable to the wide spectrum fluorescence range of coumarin-6.

Micelles morphology and size. The micelle samples were examined and imaged using a transmission electron microscope (TEM) (JEM 1400Plus, JEOL, Japan, with charge-coupled camera – Gatan Orius SC600). For TEM analysis, 5 μl of a sample were placed on a glow-discharged carbon coated 300

mesh copper TEM grid (Ted Pella, Inc). After blotting, the samples were negatively stained with a 5 μl 2% aqueous solution of uranyl acetate, and air-dried. The surfactant concentrations chosen were all above the CMC, to ensure micelle formation. Micelles size distribution was determined by dynamic light scattering (DLS) using a Malvern Autosizer 2c (Malvern Instruments, UK). The analysis was performed at a scattering angle of 90° at a temperature of 25°C for 1 ml of each surfactant solution at a concentration of 0.5 mg ml^{-1} , which is above the CMC value for all surfactants tested.

Results and discussion

CMC is a vital parameter used to characterize surfactants and other amphiphilic molecules.²⁶ Its determination is essential for surface scientists, technology developers and drug delivery researches.²⁷ Around the CMC, the properties of the solution show sharp changes that can be detected by various methods. Formation of self-assembled micelles is demonstrated by a decrease in electrical conductivity and stability in surface tension, and an increase in turbidity and organic dye solubility.²⁸ The surface tension method is relatively reproducible and simple to perform; however, a large sample volume and special equipment are required, thus reducing feasibility in some laboratories. The conductivity method, although relatively simple to perform, is quite limited, as only ionic surfactants can be measured, since the conductivity of nonionic surfactant aqueous solution is more than two orders lower than that of ionic surfactant solution.²⁰ Moreover, this method is time consuming due to the need to clean and dry electrodes after

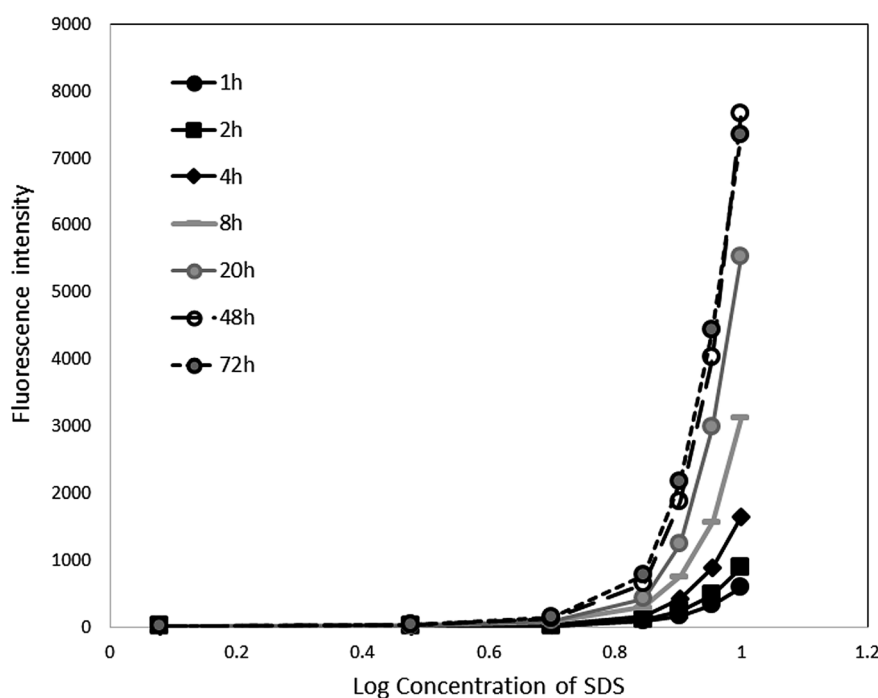


Fig. 4 Fluorescence intensity *vs.* logarithm of different concentrations of SDS. The figure shows the effect of varied rotation times from 1–72 hours on the sharpness of the exponential shape of the curve.

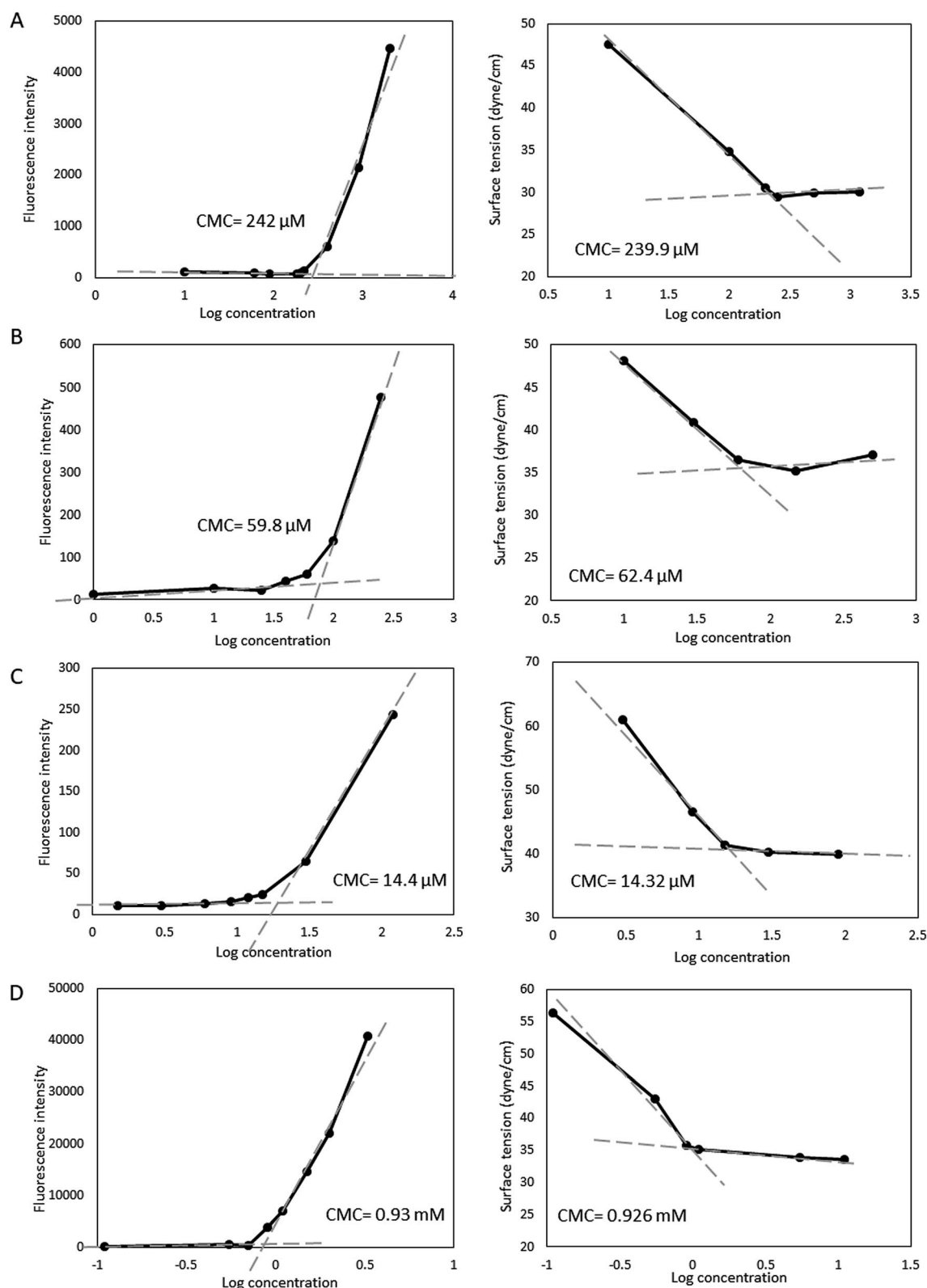


Fig. 5 Fluorescence intensity or surface tension vs. logarithm of the surfactant concentration for comparison of CMC determination methods: surface tension method (right) and coumarin-6 for dye micellization method (left). The plots represent one repeat of three for surfactants (A) Triton X100 (B) Tween 20 (C) Tween 80 and (D) CTAB. CMC was determined by the intersection of the two tangents created in the graph.

each measurement, and specially designated equipment is required as well. In both the surface tension method and the conductivity method, low sensitivity of the assay represents a central drawback, and these techniques are more appropriate for measuring CMC values above 0.1 mM.¹¹ Alternatively, the fluorescent probe or dye micellization method for CMC detection is widely used and relies on a change in fluorescence intensity emission from the dye after micelle self-assembly.¹⁷ The fluorescent probes can be classified into two types. Fluorescent turn-off probes show almost no emission in solutions but strong emission in micelles, and fluorescent turn-on probes show strong fluorescence in solution but very weak emission in micelles.²⁹ Dyes such as sudan, coumarin, rhodamine, pyrene and eosin are all suitable as turn-off probes due to a shift in the wavelength maximum when increasing the surfactant concentration, and consequently, an increase in dye solubility.¹⁷ However, these peak shift methods, like in the case of the most commonly used pyrene assay, are limited by inaccuracy and inadequate reproducibility, as discussed below.

Coumarin-6 as a common model molecule is highly hydrophobic due to the aromatic base in which presence in most coumarins, and the benzothiazolyl group and has a log *P* of 5.6 (Fig. 3A).^{30,31} The fluorescence spectrum of coumarin-6 was determined in organic solvent dimethyl-sulfoxide (DMSO) and in DDW in the presence of different surfactant solutions in concentration above the reported CMC (0.5 mg mL⁻¹). The highest fluorescence intensities of excitation and emission for coumarin-6 was 465 nm and 508 nm respectively, which are in the range of the standard filters provided by the plate reader. Emission of the probe in DMSO solution is red shifted to 515 nm probably due to a change in the dipole moment of the probe, compared to an a-polar or slightly polar environment around the probe after surfactant addition^{32,33} (Fig. 3B). The self-assembled micelles encapsulate the fluorescent probe coumarin-6 resulting in a visible change in the hydrodynamic radii of the micelle (Fig. 1). DLS results shows a clear two population distribution of micelles size, one is around 10 nm or less representing the micelles which did not encapsulate the fluorescent probe, and another more scattered and polydisperse population in the 100–300 nm range, representing the micelles that did encapsulate (Fig. 3C). The sizes of micelles measured match the sizes reported for empty surfactant micelles in the literature and the TEM images of coumarin-6 loaded micelles obtained (Fig. 1).³⁴

To achieve a robust technique that can replace current methods, we aimed to develop a simple protocol that can be

suitable also for low sample volume, that will be less time consuming than other methods and that may be adjusted for high throughput analysis. To select the minimal optimal timing of sample processing, we first calibrated the duration of rotation times needed for incubation. For the calibration process, we selected SDS as a model surfactant and examined differences in curve formation as a factor of rotation time. As shown in Fig. 4, all curves are sigmoidal and fluorescence intensity increases above the CMC. Up to 4 hours of rotation time, the intersection point in the graph is harder to detect and the value extracted is lower than previously reported.^{11,12,14} Extending the rotation time above 20 hours does not change the point of intersection significantly and as shown, after 48 hours of rotation, the curves were stable and the change in the slope angle was minimal. Of note, the increase in the fluorescence intensity above the CMC after 8 hours is high enough to calculate the intersection of the curves but might not be suitable for surfactants that need longer time to reach equilibrium and an exponentially shaped curve. Therefore, we conclude that overnight rotation is more convenient and accurate and may be more universally used.

The Du Noüy ring method is one of the more commonly used techniques to measure CMC. Accordingly, surface tension is measured as a function of increasing surfactant concentrations. At the point at which the surface is saturated with surfactant, micelles are formed and the surface tension remains constant.¹ Although this is a standard method, there are some disadvantages that will be discussed below. We compared the results of coumarin-6 encapsulations inside different surfactant types (cationic, anionic and nonionic), as obtained by our new protocol, to the values obtained in surface tension measurements by the du Noüy ring method. As shown in Fig. 5, concentrations ranging from 0 to values high above the known CMC point were tested in both methods and the values were plotted against the concentration to estimate the CMC. CTAB was selected as an example of cationic surfactant and measured in concentrations of 0–10 mM. Tween 20, Tween 80 and Triton X100, as nonionic surfactants, were measured in concentrations of 0–0.25 mM, 0–0.1 mM and 0–1.2 mM, respectively. Tangent lines were drawn for points below and above the breaking stage, and the CMC was determined as the point at which those two lines intercept. Table 1 summarizes CMC values as measured using the two methods compared to values stated in the literature. A slightly higher difference in the value obtained by the two methods for non-ionic surfactant is well known in the

Table 1 CMC values (in μM) for Triton X100, Tween 80 and 20, CTAB and SDS, as obtained from our studies compared with the surface tension method and values reported in the literature

Surfactant	Fluorescent dye	Surface tension	Literature	Ref.
Triton X100	239.5 (± 6.6)	236.8 (± 9.3)	190–270	11, 14, 16, 19 and 29
Tween 80	14.8 (± 0.6)	14.3 (± 0.4)	12–15	28, 30 and 31
Tween 20	58.6 (± 3.5)	62.9 (± 6.6)	48–60	19, 28, 30 and 31
CTAB	0.9×10^3 (± 0.08)	0.93×10^3 (± 0.005)	0.8 to 1×10^3	11, 16, 19 and 32
SDS	7.9×10^3 (± 0.1)	8.03×10^3 (± 0.03)	7.9 to 8.2×10^3	11, 12, 14, 16 and 19

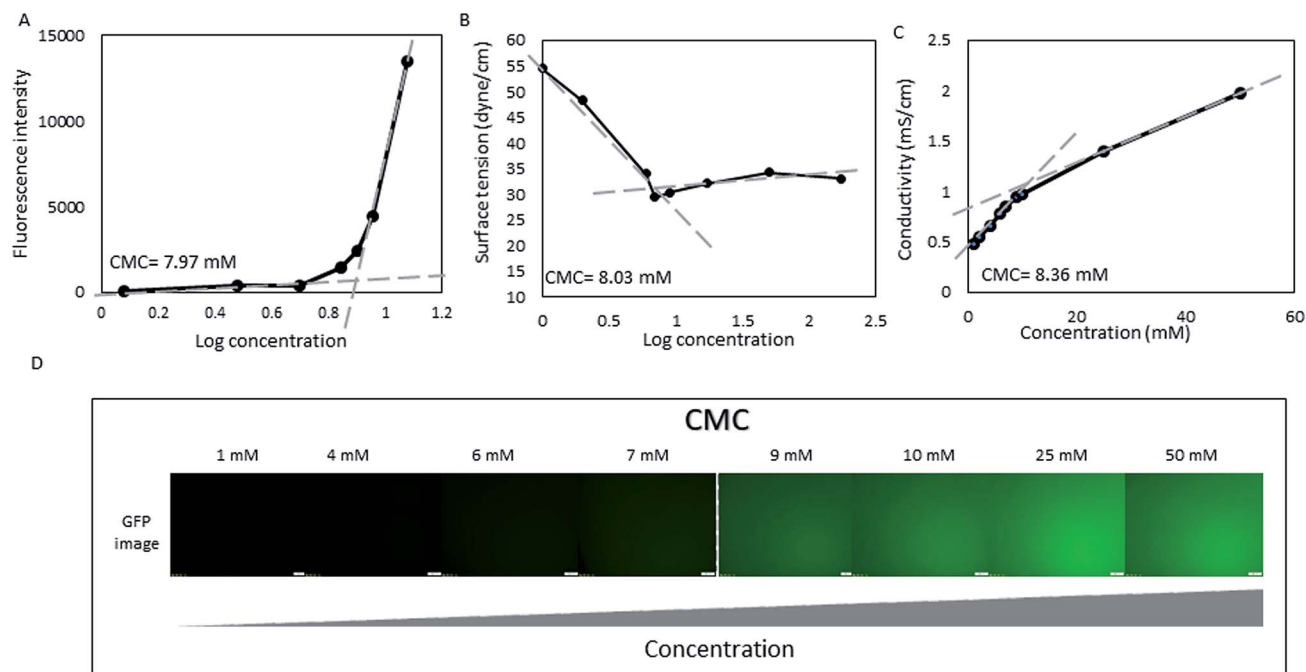


Fig. 6 CMC of SDS by different methods of determination. The CMC evaluated by (A) the coumarin-6 dye micellization method by examining fluorescence intensity, (B) the surface tension method by the duNouy ring measurements, (C) the conductivity method by using conductometer and (D) visualized by examining the fluorescent light emitted by fluorescent microscopy. The plots represent one repeat out of three.

literature and caused by the presence of impurities and the broad molecular weight distribution.³⁵ Overall, the results obtained with the two methods were similar to the values reported in the literature; this supports the reliability of the new method.

We further compared the CMC of SDS to other available determination methods, such as the conductivity method, which is a well-established technique for CMC determination. Using a conductometer, the conductivity of the solution can be calculated from the intersection point of curves before and after the breaking point of the graph, the same as for other methods described. The CMC value obtained from this method was 8.36 mM, which is less accurate than the value obtained when the methods are compared; this highlights the low sensitivity of the method. The CMC of SDS can be visually detected, thus exploiting the fluorescence of the coumarin-6. Above the CMC, the dye located inside the micelles emits fluorescent light detected by a GFP filter (Fig. 6), at concentrations above 9 mM, compared with the concentration set below 7 mM, which emits no light.

In contrast to the surface tension and conductivity methods, in which detection is performed indirectly, the use of probes to detect the point at which the micelles first form enables direct detection. Surface tension and conductivity methods measure the surface concentration of face active species and the electrolyte activity of the monomers respectively, and not the presence of micelles in the solution.^{27,35} Pyrene as a fluorescent dye for CMC assessment is usually used as a probe by following the I_1/I_3 method, which refers to the ratio between the first and third vibronic peaks at 374 nm and 483 nm, respectively, when excitation occurs at 336 nm.¹⁷ The fluorescence spectrum

emitted is measured by a spectrophotometer that requires a quartz cuvette and is further analyzed for CMC calculation.

The pyrene method suffers from low reproducibility since the signal peaks of pyrene-supplemented samples are often shifted, necessitating the performance of a spectral screen for each and every sample at every measured concentration, followed by ratio calculations.¹⁶ In addition, the pyrene assay requires the use of fluorometers with multiwavelength capability, which is not a standard filter-based plate reader thus creating a big difference in the fluorescence spectrum of pyrene at the same concentration of a surfactant. Distinguishing between the vibronic peaks of pyrene is obvious when examining the fluorescence spectrum with a spectrofluorometer, but hard to detect when examining with a plate reader (SDS example ESI 2†). The result is substantially limited accessibility to this method. It is quite surprising that such an old and laborious technique is still so commonly used and has not yet been replaced by equivalent but much simpler assays. The ability to read a single fluorescent value using a standard plate reader may produce much quicker results that are also suitable for high throughput.²⁹ Also other fluorescent probes that have been used for CMC determination have yielded results that are not sufficiently accurate compared with the values accepted in the literature or have vast standard deviation, for example, using other coumarins like coumarin-153 for SDS³² or Nile red dye for CTAB.¹¹

Conclusions

The CMC values may fluctuate according to the method applied, surfactant structural properties, ionic strength, solution

impurities and the fluorescent probe chosen.¹¹ The values achieved by the novel method described concur with values reported in the literature for all surfactant tested. Several methods exist for determining the CMC and appropriate dyes for the dye micellization method, but not one universally agreed and reproducible technique has been proposed.^{11,17,30} Coumarin-6 demonstrated compatibility as a fluorescent probe since it was encapsulated at low concentrations by the micelles above the CMC. The method is easy to perform and yielded accurate and reproducible CMC values. Therefore, the method presented here is more universal and can be applicable for cationic, anionic or nonionic surfactants. We believe that adapting such method would provide a clear benefit for the pharmaceutical field.

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Conflicts of interest

The authors declare no conflicts of interest.

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